

MMDB - Ex 05

1.1 $P_{\text{output}} = \alpha P_{\text{input}} + \beta$, 8-bit grayscale ($P \in [0, 255]$)
 α = Contrast adjustment

β = Brightness adjustment

1.2. Image inversion - For 8-bit $\rightarrow (255 - P)$

$$255 - P = \alpha P + \beta$$

$$\alpha = -1$$

$$\beta = 255$$

1.3. If the α and β parameters results in P_{out} below 0 or above 255 the raw result is an undefined 8-bit image.

Fix: Clamping, Saturate any value below 0 to 0 and above 255 to 255

1.4 Contrast adjustment $\rightarrow f(p) = \alpha \cdot p$

$$\alpha a + \beta = 0$$

$$\alpha b + \beta = 255$$

$$\alpha b + \beta = 255$$

$$\alpha a + \beta = 0$$

$$\alpha b - \alpha a = 255$$

$$\alpha(b - a) = 255$$

$$\alpha = \frac{255}{b - a}$$

$$\beta = -\frac{\alpha \cdot a}{(b - a)}$$

$$\beta = -\frac{255 \cdot a}{(b - a)}$$

2.1 The kernel can extend beyond the image boundary at the edges

Fixes = ① Redefines the convolution when it touches the boundary

② Pad the image first (fills missing cells with 0 or reflect the values across the edges)

2.2. $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \times \left(\frac{1}{25}\right)$

2.3 Weighted - smoothing kernels ($\alpha = 2$):

$$f(x, y) = -2 \max(|x|, |y|) + 4$$

$$(K=4)$$

Distance $\max(|x|, |y|)$ For offsets

(0,0) ... ($\pm 2, \pm 2$)

Kernel: $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 2 & 2 & 0 \end{bmatrix}$

Conrad: $f(x, y) = -2\sqrt{x^2 + y^2} + 4$

$$\begin{bmatrix} 0 & 2 & 4 & 2 & 0 \\ 0 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2.4. Laplacian filters are derivative filters used to find edges in images

Example:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

or

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$